# Standard sheaf-cohomology vanishing on $\mathbb{CP}^2$ and the Picard classification of holomorphic line bundles: a background auxiliary note for $\mathbb{CP}^2$-based versions of the TECT Pillar-4 bundle programme

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We collect, in a self-contained presentation, two standard algebro-geometric facts on the complex projective surface $\mathbb{CP}^2$: (A) the vanishing of the holomorphic structure-sheaf cohomology groups $H^1(\mathbb{CP}^2, \mathcal{O}) = 0$ and $H^2(\mathbb{CP}^2, \mathcal{O}) = 0$, derived via Hodge theory (Dolbeault cohomology / Hodge diamond); (B) the resulting discrete Picard classification $\mathrm{Pic}(\mathbb{CP}^2) \cong \mathbb{Z}$, $\mathrm{Pic}^0(\mathbb{CP}^2) = 0$, so that every holomorphic line bundle on $\mathbb{CP}^2$ is uniquely determined by its integer first Chern class $k \in \mathbb{Z}$ with no continuous moduli. Two independent derivations are presented (Hodge-theoretic and Čech-cohomological). These are standard textbook facts; the present note collects them as a useful geometric background for earlier $\mathbb{CP}^2$-based versions of the TECT Pillar-4 bundle programme, but they do NOT by themselves resolve the current $\Sigma_0$-based realisation chain (the canonical TECT Pillar-4 realisation programme uses $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305, not $\mathbb{CP}^2$, and the proof-rigor / realisation-gate carve-out per Math302 keeps the two layers separately tracked). The novelty of the present note is low; its role is auxiliary / expository.

## I. INTRODUCTION

Complex projective two-space $\mathbb{CP}^2$ is a fundamental object in algebraic and differential geometry, arising naturally as:

1. The space of complex lines through the origin in $\mathbb{C}^3$.

2. The space of complex projectivised divisors on $\mathbb{CP}^1 \times \mathbb{CP}^1$.

3. A base space for vector-bundle constructions in gauge theory and physics.

In topological energy condensate theory (TECT), *earlier versions* of the Pillar 4 bundle programme used a $\mathbb{CP}^2$-type base space for the classification of $\mathrm{U}(1)_\chi$ line bundles. For those earlier versions, the vanishing of $H^1(\mathbb{CP}^2, \mathcal{O})$ implies that no hidden continuous moduli appear, and the

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topological classification via the integer first Chern class is exhaustive. The *current* canonical Pillar 4 realisation programme operates on the base $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305 with the proof-rigor / realisation-gate carve-out of Math302; the $\mathbb{CP}^2$ statements collected here are therefore an auxiliary background note for the earlier $\mathbb{CP}^2$-based versions, not a current canonical Pillar 4 closure claim.

## II. SETTING AND MAIN RESULT

### A. Complex projective two-space

$\mathbb{CP}^2$ is defined as:

$$\mathbb{CP}^2 = \{[z_0 : z_1 : z_2] : (z_0, z_1, z_2) \in \mathbb{C}^3 \setminus \{0\}\}/\mathbb{C}^*. \tag{1}$$

As a smooth complex manifold, it has complex dimension 2 (real dimension 4). The holomorphic structure sheaf $\mathcal{O}$ assigns to each open set $U$ the ring of holomorphic functions on $U$. Line bundles are classified by the Picard group $\mathrm{Pic}(\mathbb{CP}^2)$, the group of isomorphism classes of holomorphic line bundles.

### B. Hodge diamond of $\mathbb{CP}^2$

For a compact complex manifold, the Hodge diamond $h^{p,q}$ encodes the dimensions of holomorphic differential forms. For $\mathbb{CP}^2$:

$$\begin{array}{ccccc} & & h^{0,0} = 1 & & \\ & h^{0,1} = 0 & & h^{1,0} = 0 & \\ h^{0,2} = 0 & & h^{1,1} = 1 & & h^{2,0} = 0 \\ & h^{1,2} = 0 & & h^{2,1} = 0 & \\ & & h^{2,2} = 1 & & \end{array} \tag{2}$$

### C. Main theorem

**Theorem 1** (Sheaf cohomology vanishing and Picard classification)**.** *Let $\mathbb{CP}^2$ be complex projective two-space with its standard holomorphic structure.*

**(A)** *The sheaf-cohomology groups of the structure sheaf vanish:*

$$H^0(\mathbb{CP}^2, \mathcal{O}) = \mathbb{C}, \quad H^1(\mathbb{CP}^2, \mathcal{O}) = 0, \quad H^2(\mathbb{CP}^2, \mathcal{O}) = 0. \tag{3}$$

**(B)** *The Picard group is isomorphic to the integers:*

$$\boxed{\mathrm{Pic}(\mathbb{CP}^2) \cong \mathbb{Z}.} \tag{4}$$

*Every holomorphic line bundle $\mathcal{L}$ on $\mathbb{CP}^2$ is isomorphic to $\mathcal{O}(k)$ for a unique $k \in \mathbb{Z}$, where $\mathcal{O}(1)$ is the hyperplane bundle. The isomorphism is given by the first Chern class:*

$$c_1 : \mathrm{Pic}(\mathbb{CP}^2) \to H^2(\mathbb{CP}^2, \mathbb{Z}) \cong \mathbb{Z}. \tag{5}$$

*In particular,* $\mathrm{Pic}^0(\mathbb{CP}^2) = \{$*trivial bundle*$\}$ *(the connected component of the identity has no continuous deformations).*

## III. PROOF SKETCH

*a. Step 1 (Hodge-theoretic proof of vanishing).* By Dolbeault's theorem, for a compact Kähler manifold $X$,

$$H^q(X, \Omega^p) \cong H^{p,q}(X), \tag{6}$$

where $H^{p,q}$ is the Dolbeault cohomology group of $(p, q)$-forms.

For $p = 0$ (functions), $\Omega^0 = \mathcal{O}$, so:

$$H^q(\mathbb{CP}^2, \mathcal{O}) = H^{0,q}(\mathbb{CP}^2) = h^{0,q}(\mathbb{CP}^2). \tag{7}$$

Reading the Hodge diamond, $h^{0,1} = 0$ and $h^{0,2} = 0$, giving:

$$H^1(\mathbb{CP}^2, \mathcal{O}) = 0, \quad H^2(\mathbb{CP}^2, \mathcal{O}) = 0. \tag{8}$$

*b. Step 2 (Picard-group classification via exponential sequence).* The exponential sheaf sequence

$$0 \to \mathbb{Z} \xrightarrow{2\pi i} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \to 0 \tag{9}$$

gives the long exact sequence in cohomology:

$$\cdots \to H^1(\mathbb{CP}^2, \mathcal{O}) \to H^1(\mathbb{CP}^2, \mathcal{O}^*) \xrightarrow{c_1} H^2(\mathbb{CP}^2, \mathbb{Z}) \to H^2(\mathbb{CP}^2, \mathcal{O}) \to \cdots . \tag{10}$$

Since $H^1(\mathbb{CP}^2, \mathcal{O}) = 0$ and $H^2(\mathbb{CP}^2, \mathcal{O}) = 0$, this sequence becomes:

$$0 \to H^1(\mathbb{CP}^2, \mathcal{O}^*) \xrightarrow{c_1} H^2(\mathbb{CP}^2, \mathbb{Z}) \to 0. \tag{11}$$

The map $c_1$ is an isomorphism. Since $H^1(\mathbb{CP}^2, \mathcal{O}^*) \cong \mathrm{Pic}(\mathbb{CP}^2)$ (by standard sheaf cohomology), we have:

$$\mathrm{Pic}(\mathbb{CP}^2) \cong H^2(\mathbb{CP}^2, \mathbb{Z}). \tag{12}$$

Now, $H^2(\mathbb{CP}^2, \mathbb{Z}) \cong \mathbb{Z}$ (generated by the hyperplane class $H$), so:

$$\mathrm{Pic}(\mathbb{CP}^2) \cong \mathbb{Z}. \tag{13}$$

*c. Step 3 (Consequence: no continuous Pic$^0$).* The Picard group decomposes as $\mathrm{Pic}(X) = \mathrm{Pic}^0(X) \times \mathrm{NS}(X)$, where $\mathrm{NS}(X)$ is the Neron–Severi group (the image of $c_1$, a finitely-generated free abelian group) and $\mathrm{Pic}^0(X)$ is the connected component of the identity (the "Picard variety").

For $\mathbb{CP}^2$, since $\mathrm{Pic}(\mathbb{CP}^2) \cong \mathbb{Z}$ is discrete, $\mathrm{Pic}^0(\mathbb{CP}^2) = \{0\}$. There is no continuous deformation of line bundles.

## IV. AUXILIARY APPLICATION TO $\mathbb{CP}^2$-BASED VERSIONS OF THE TECT PILLAR-4 PROGRAMME

The current canonical TECT Pillar-4 realisation programme operates on the base $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305, NOT on $\mathbb{CP}^2$. Earlier versions of the Pillar-4 bundle programme used a $\mathbb{CP}^2$-type base. The statement below is phrased as an if-then auxiliary corollary for those earlier $\mathbb{CP}^2$-based versions; it is NOT a current canonical Pillar-4 closure claim, and the canonical bookkeeping (Math302 proof-rigor / realisation-gate carve-out) requires that the two layers be tracked separately.

**Corollary 2** ($\mathbb{CP}^2$-base auxiliary statement, NOT a current Pillar-4 closure)**.** *If a TECT realisation is formulated on a $\mathbb{CP}^2$-type base and the $U(1)_\chi$ sector is represented by a holomorphic line bundle on $\mathbb{CP}^2$, then that line bundle is classified discretely by an integer first Chern class $k_\chi \in \mathbb{Z}$, with no continuous* Pic$^0$ *modulus. The holomorphic and smooth bundle structures coincide on their topological data. This is an immediate consequence of Theorem 1(B) and is independent of any TECT-specific input.*

## V. DISCUSSION

The discrete Picard classification on $\mathbb{CP}^2$ records that holomorphic line bundles do not introduce continuous moduli on this base. As emphasised in the abstract and in Corollary 2, this is auxiliary input for earlier $\mathbb{CP}^2$-based versions of the Pillar-4 programme; the current canonical realisation programme operates on $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305, where a separate (and not derived in this note) cohomological calculation applies. The mathematics presented here is standard; the role of this paper is expository / background-collecting rather than novel-result-claiming.

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